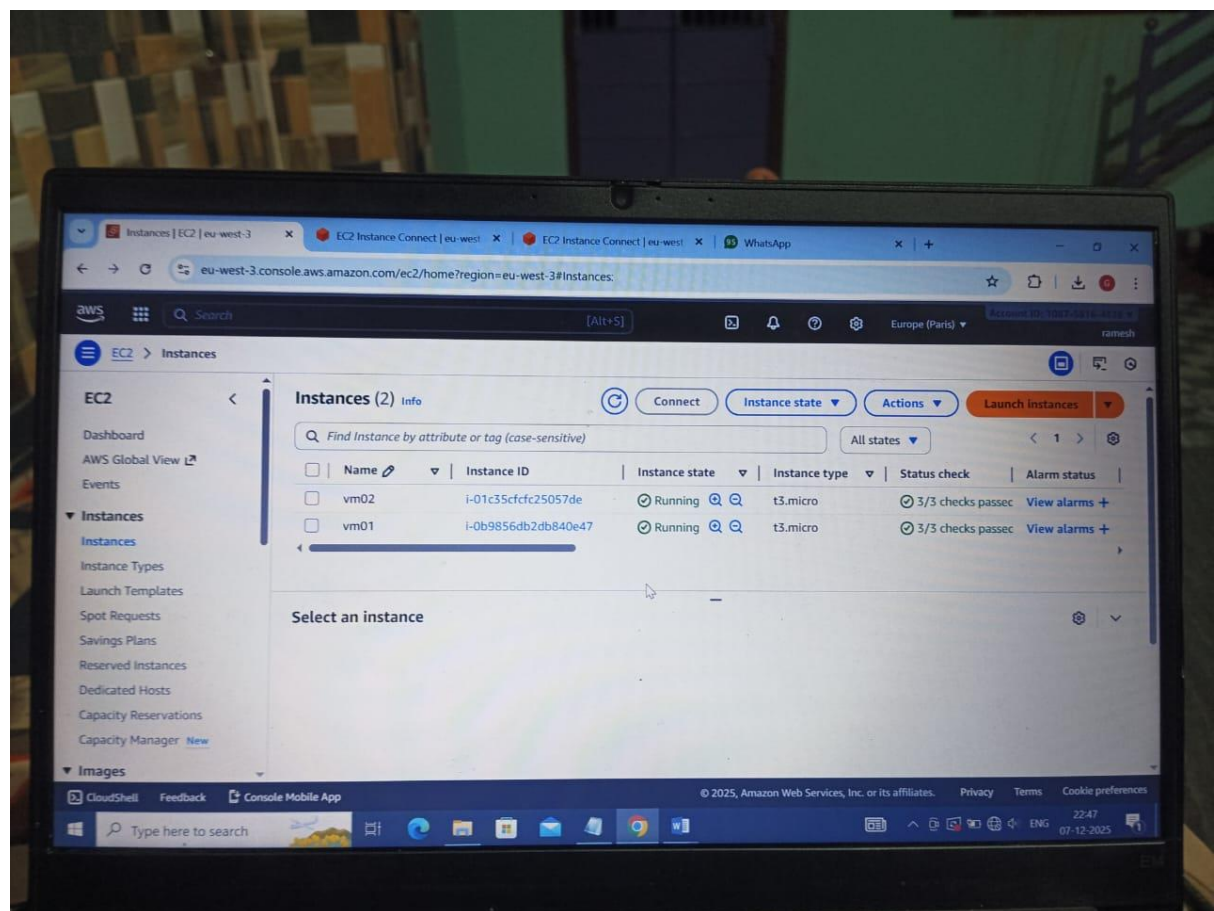


Tasks:

1. Initial Setup and Directory Structure (Instance 1)

- Launch two EC2 instances (Instance 1 and Instance 2) in the same region using a Linux AMI (e.g., Ubuntu 20.04 or Amazon Linux 2).
- Ensure both instances can communicate with each other by configuring the security groups to allow SSH
- SSH into Instance 1 and create a directory structure in your home directory called `project_setup` with the following subdirectories:
 - `data`
 - `backups`
 - `logs`
 - `temp`
 - `scripts`
- Launch two EC2 instances (Instance 1 and Instance 2) in the same region using a Linux AMI (e.g., Ubuntu 20.04 or Amazon Linux 2).

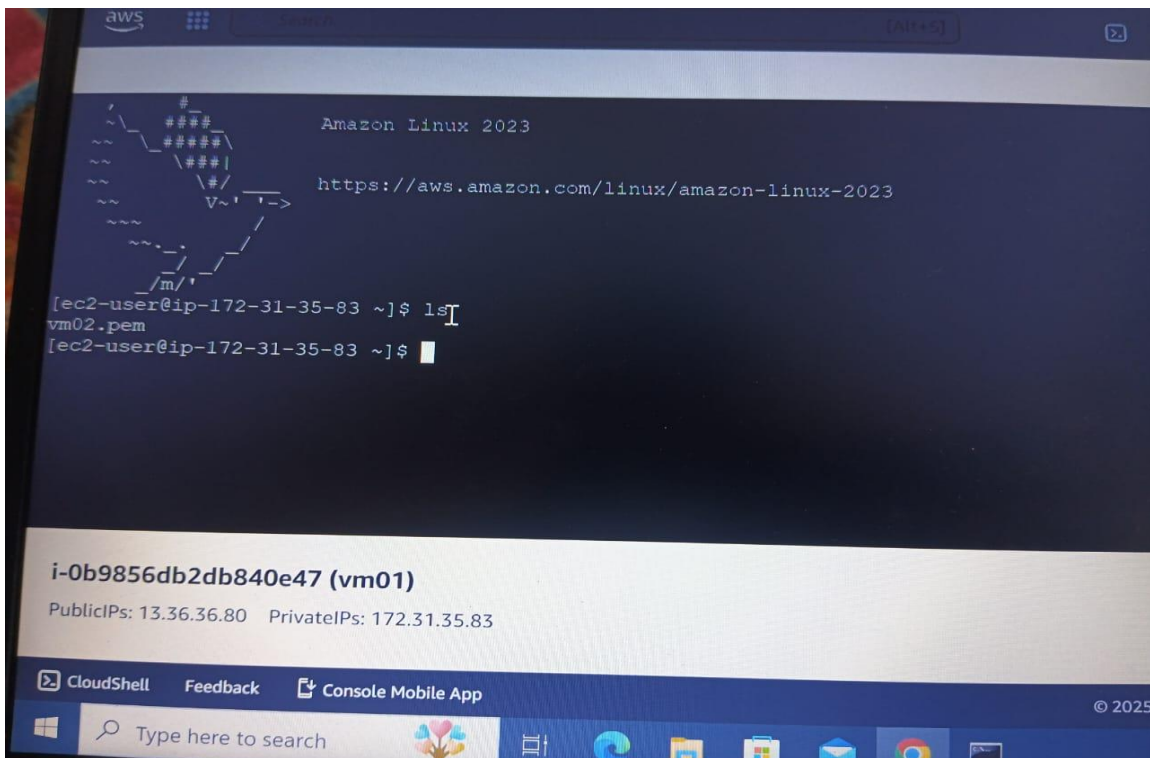


- ```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.19045.2965]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Lenovo\Downloads>scp -i vm01.pem vm02.pem ec2-user@13.36.36.80:/home/ec2-user
The authenticity of host '13.36.36.80 (13.36.36.80)' can't be established.
ECDSA key fingerprint is SHA256:0eOyX7VptfTstjbmw4ye7fAJmAcMv65NavSD8P7mQew.
Are you sure you want to continue connecting (yes/no/[fingerprint])?
lost connection

C:\Users\Lenovo\Downloads>scp -i vm01.pem vm02.pem ec2-user@13.36.36.80:/home/ec2-user
The authenticity of host '13.36.36.80 (13.36.36.80)' can't be established.
ECDSA key fingerprint is SHA256:0eOyX7VptfTstjbmw4ye7fAJmAcMv65NavSD8P7mQew.
Are you sure you want to continue connecting (yes/no/[fingerprint])?
Warning: Permanently added '13.36.36.80' (ECDSA) to the list of known hosts.
vm02.pem

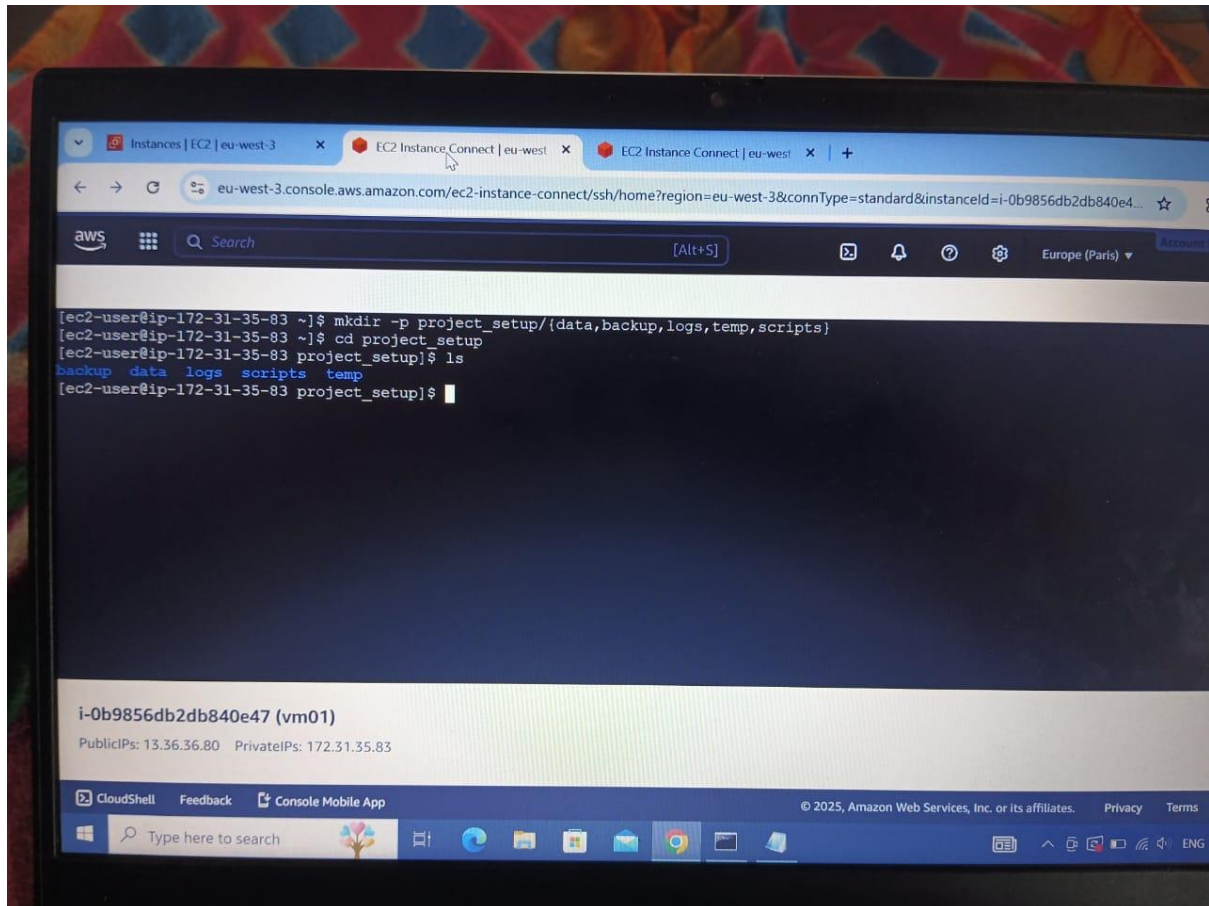
C:\Users\Lenovo\Downloads>scp -i vm01.pem vm02.pem ec2-user@51.44.6.138:/home/ec2-user
The authenticity of host '51.44.6.138 (51.44.6.138)' can't be established.
```





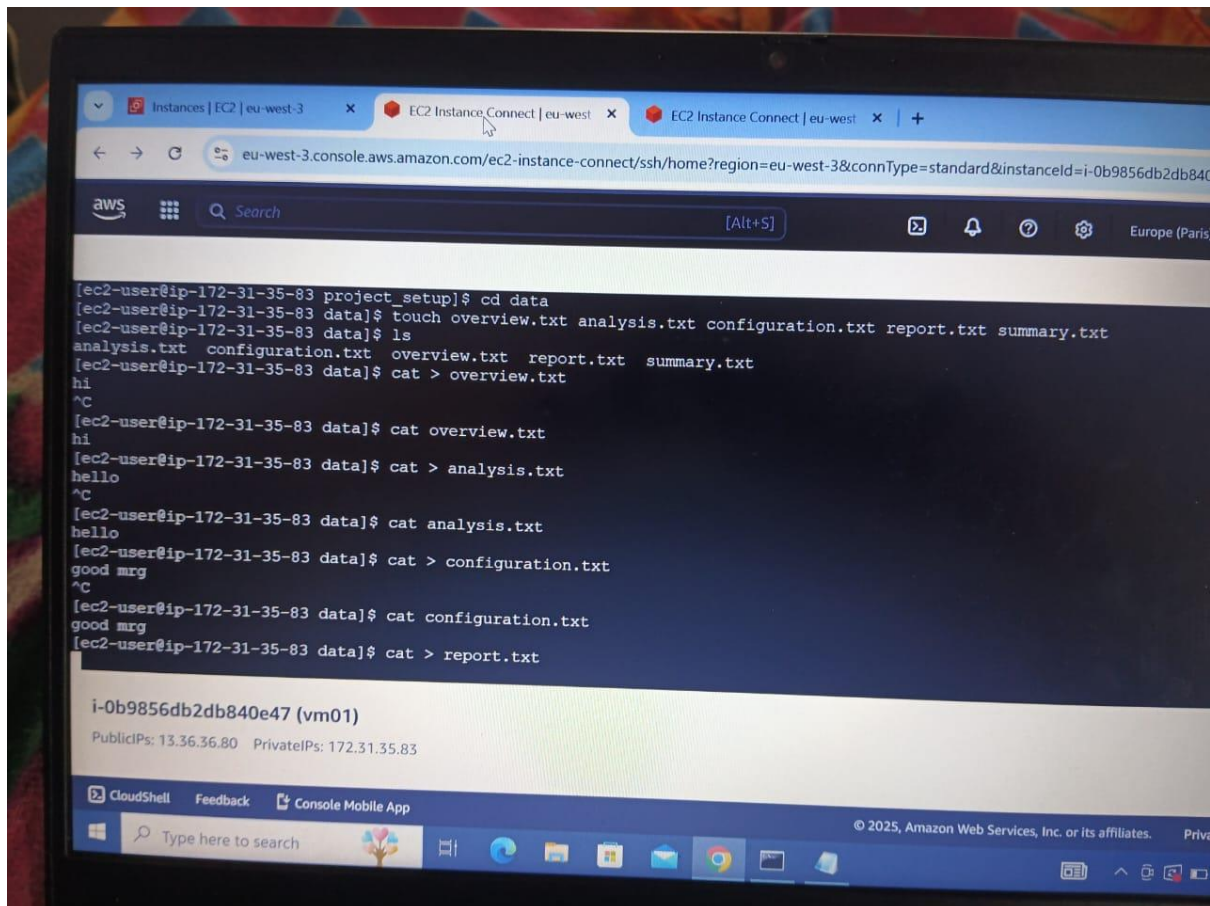


- SSH into Instance 1 and create a directory structure in your home directory called project\_setup with the following subdirectories:
  - data
  - backups
  - logs
  - temp
  - scripts



## 2. File Creation and Content Population (Instance 1)

- Inside the data directory, create five text files named:
  - overview.txt
  - analysis.txt
  - configurations.txt
  - report.txt
  - summary.txt
- Populate each file with content using the echo or cat commands.



### 3. File Operations and Management (Instance 1)

- Copy overview.txt from the data directory to the backups directory.
- Move report.txt from data to logs.
- Rename analysis.txt to analysis\_backup.txt in the data directory.
- Create a new file archive.txt in the data directory.
- Append content to archive.txt using output redirection (e.g., echo "Backup process started." >> archive.txt).

```
aws console.aws.amazon.com/ec2-instance-connect/ssh/home?region=eu-west-3&connType=standard&instanceId=i-0b9856db2db840e47

[ec2-user@ip-172-31-35-83 ~]$ cd project_setup
[ec2-user@ip-172-31-35-83 project_setup]$ ls
backup data logs scripts temp
[ec2-user@ip-172-31-35-83 project_setup]$ cd data
[ec2-user@ip-172-31-35-83 data]$ ls
analysis.txt configuration.txt overview.txt report.txt summary.txt
[ec2-user@ip-172-31-35-83 data]$ cp overview.txt /home/ec2-user/project_setup/backup
[ec2-user@ip-172-31-35-83 project_setup]$ cd backup
[ec2-user@ip-172-31-35-83 backup]$ ls
overview.txt
[ec2-user@ip-172-31-35-83 backup]$ cd ..
[ec2-user@ip-172-31-35-83 project_setup]$ cd data
[ec2-user@ip-172-31-35-83 data]$ ls
analysis.txt configuration.txt overview.txt report.txt summary.txt
[ec2-user@ip-172-31-35-83 data]$ mv report.txt /home/ec2-user/project_setup/logs
[ec2-user@ip-172-31-35-83 data]$ cd ..
[ec2-user@ip-172-31-35-83 project_setup]$ cd logs
[ec2-user@ip-172-31-35-83 logs]$ ls
report.txt
[ec2-user@ip-172-31-35-83 logs]$

i-0b9856db2db840e47 (vm01)
PublicIPs: 13.36.36.80 PrivateIPs: 172.31.35.83

CloudShell Feedback Console Mobile App
Type here to search
```

```
Instances | EC2 | eu-west-3 | EC2 Instance Connect | eu-west-3 | EC2 Instance Connect | eu-west-3 | +
eu-west-3.console.aws.amazon.com/ec2-instance-connect/ssh/home?region=eu-west-3&connType=standard&instanceId=i-0b9856db2db840e47

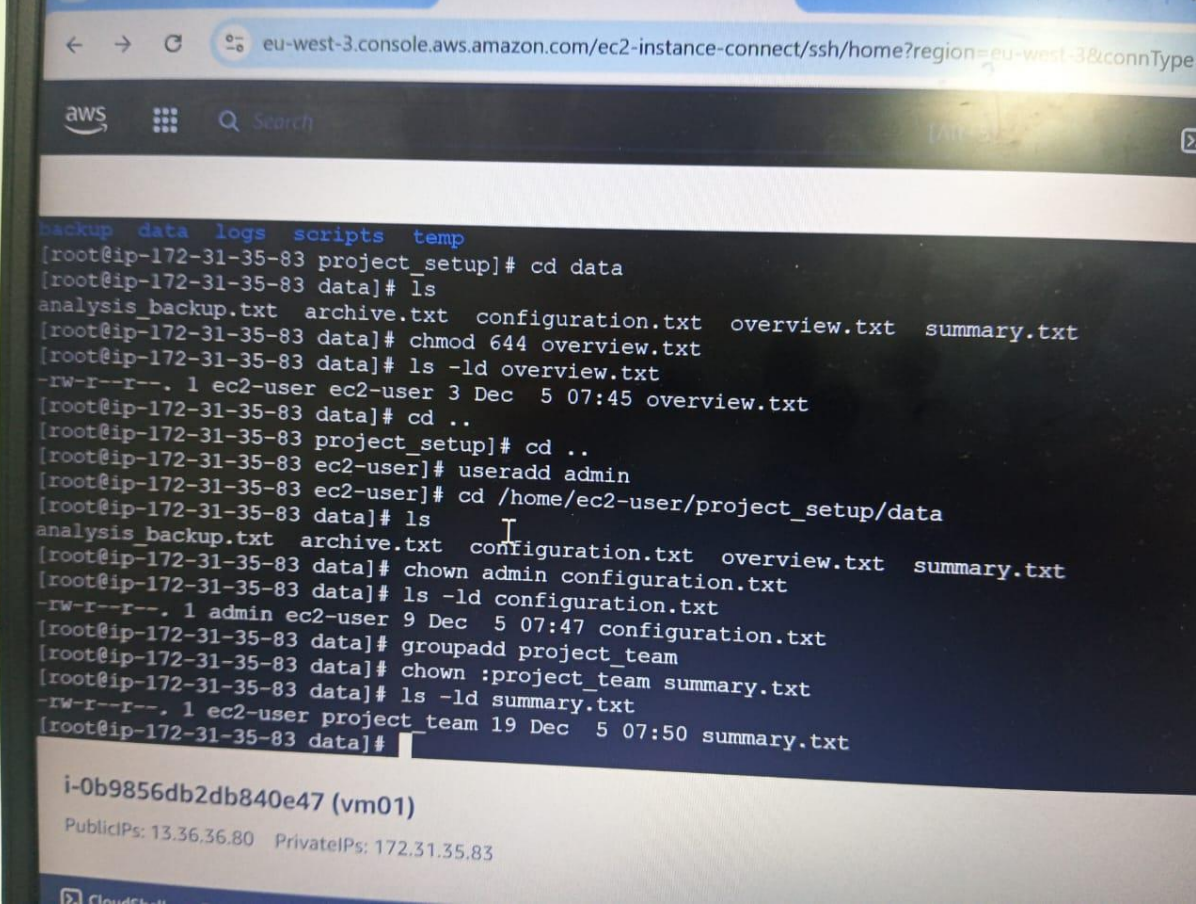
[ec2-user@ip-172-31-35-83 data]$ ls
analysis.txt configuration.txt overview.txt report.txt summary.txt
[ec2-user@ip-172-31-35-83 data]$ mv report.txt /home/ec2-user/project_setup/logs
[ec2-user@ip-172-31-35-83 data]$ cd ..
[ec2-user@ip-172-31-35-83 project_setup]$ cd logs
[ec2-user@ip-172-31-35-83 logs]$ ls
report.txt
[ec2-user@ip-172-31-35-83 logs]$ cd ..
[ec2-user@ip-172-31-35-83 project_setup]$ cd data
[ec2-user@ip-172-31-35-83 data]$ ls
analysis.txt configuration.txt overview.txt summary.txt
[ec2-user@ip-172-31-35-83 data]$ mv analysis.txt analysis_backup.txt
[ec2-user@ip-172-31-35-83 data]$ ls
analysis_backup.txt configuration.txt overview.txt summary.txt
[ec2-user@ip-172-31-35-83 data]$ touch archive.txt
[ec2-user@ip-172-31-35-83 data]$ ls
analysis_backup.txt archive.txt configuration.txt overview.txt summary.txt
[ec2-user@ip-172-31-35-83 data]$ echo "good morning" >> archive.txt
[ec2-user@ip-172-31-35-83 data]$ cat archive.txt
good morning
[ec2-user@ip-172-31-35-83 data]$

i-0b9856db2db840e47 (vm01)
PublicIPs: 13.36.36.80 PrivateIPs: 172.31.35.83
```



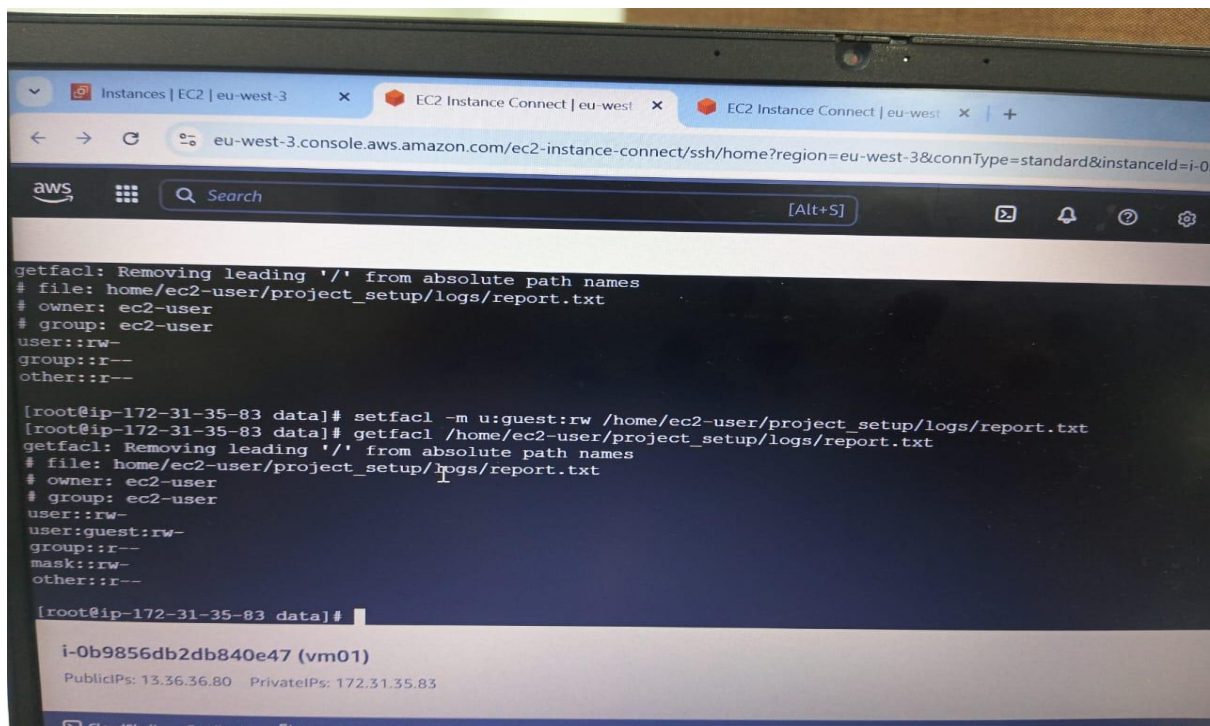
#### 4. File Permissions and Ownership (Instance 1)

- Set the permissions of overview.txt to be readable and writable by the owner, and readable by others.
- Change the ownership of configurations.txt to a user named admin.
- Set the group ownership of summary.txt to a group named project\_team.
- Set ACLs so that a user named guest has read and write permissions on report.txt.
- Remove ACL entries from report.txt.
- Use the ls -l and getfacl commands to verify the permissions and ownership.



```
eu-west-3.console.aws.amazon.com/ec2-instance-connect/ssh/home?region=eu-west-3&connType=
aws Search [Alt-3]
[backup_data logs scripts temp]
[root@ip-172-31-35-83 project_setup]# cd data
[root@ip-172-31-35-83 data]# ls
analysis_backup.txt archive.txt configuration.txt overview.txt summary.txt
[root@ip-172-31-35-83 data]# chmod 644 overview.txt
[root@ip-172-31-35-83 data]# ls -ld overview.txt
-rw-r--r--. 1 ec2-user ec2-user 3 Dec 5 07:45 overview.txt
[root@ip-172-31-35-83 data]# cd ..
[root@ip-172-31-35-83 project_setup]# cd ..
[root@ip-172-31-35-83 ec2-user]# useradd admin
[root@ip-172-31-35-83 ec2-user]# cd /home/ec2-user/project_setup/data
[root@ip-172-31-35-83 data]# ls
analysis_backup.txt archive.txt configuration.txt overview.txt summary.txt
[root@ip-172-31-35-83 data]# chown admin configuration.txt
[root@ip-172-31-35-83 data]# ls -ld configuration.txt
-rw-r--r--. 1 admin ec2-user 9 Dec 5 07:47 configuration.txt
[root@ip-172-31-35-83 data]# groupadd project_team
[root@ip-172-31-35-83 data]# chown :project_team summary.txt
[root@ip-172-31-35-83 data]# ls -ld summary.txt
-rw-r--r--. 1 ec2-user project_team 19 Dec 5 07:50 summary.txt
[root@ip-172-31-35-83 data]#

i-0b9856db2db840e47 (vm01)
PublicIPs: 13.36.36.80 PrivateIPs: 172.31.35.83
CloudShell
```



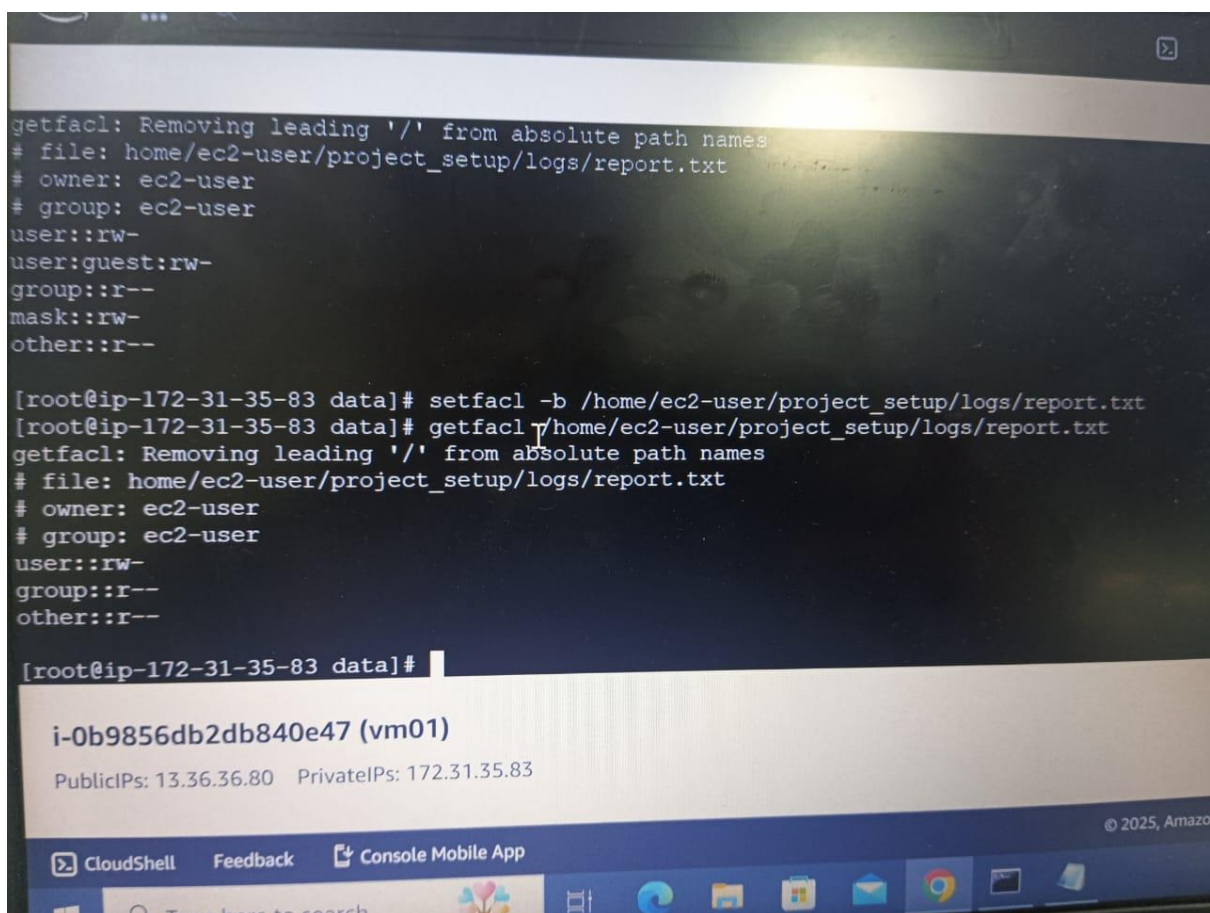
The screenshot shows the AWS CloudShell interface. The browser tabs at the top include 'Instances | EC2 | eu-west-3' and 'EC2 Instance Connect | eu-west-3'. The address bar shows the URL: eu-west-3.console.aws.amazon.com/ec2-instance-connect/ssh/home?region=eu-west-3&connType=standard&instanceId=i-0b9856db2db840e47. The terminal window displays the following commands and output:

```
getfacl: Removing leading '/' from absolute path names
file: home/ec2-user/project_setup/logs/report.txt
owner: ec2-user
group: ec2-user
user::rw-
group::r--
other::r--

[root@ip-172-31-35-83 data]# setfacl -m u:guest:rw /home/ec2-user/project_setup/logs/report.txt
[root@ip-172-31-35-83 data]# getfacl /home/ec2-user/project_setup/logs/report.txt
getfacl: Removing leading '/' from absolute path names
file: home/ec2-user/project_setup/logs/report.txt
owner: ec2-user
group: ec2-user
user::rw-
user:guest:rw-
group::r--
mask::rw-
other::r--

[root@ip-172-31-35-83 data]#
```

Below the terminal, the instance ID **i-0b9856db2db840e47 (vm01)** is shown, along with PublicIPs: 13.36.36.80 and PrivateIPs: 172.31.35.83. The bottom of the interface includes a 'CloudShell' label, a 'Feedback' link, and a 'Console Mobile App' download button.



This screenshot shows the same AWS CloudShell terminal session. The terminal output is as follows:

```
getfacl: Removing leading '/' from absolute path names
file: home/ec2-user/project_setup/logs/report.txt
owner: ec2-user
group: ec2-user
user::rw-
user:guest:rw-
group::r--
mask::rw-
other::r--

[root@ip-172-31-35-83 data]# setfacl -b /home/ec2-user/project_setup/logs/report.txt
[root@ip-172-31-35-83 data]# getfacl /home/ec2-user/project_setup/logs/report.txt
getfacl: Removing leading '/' from absolute path names
file: home/ec2-user/project_setup/logs/report.txt
owner: ec2-user
group: ec2-user
user::rw-
group::r--
other::r--

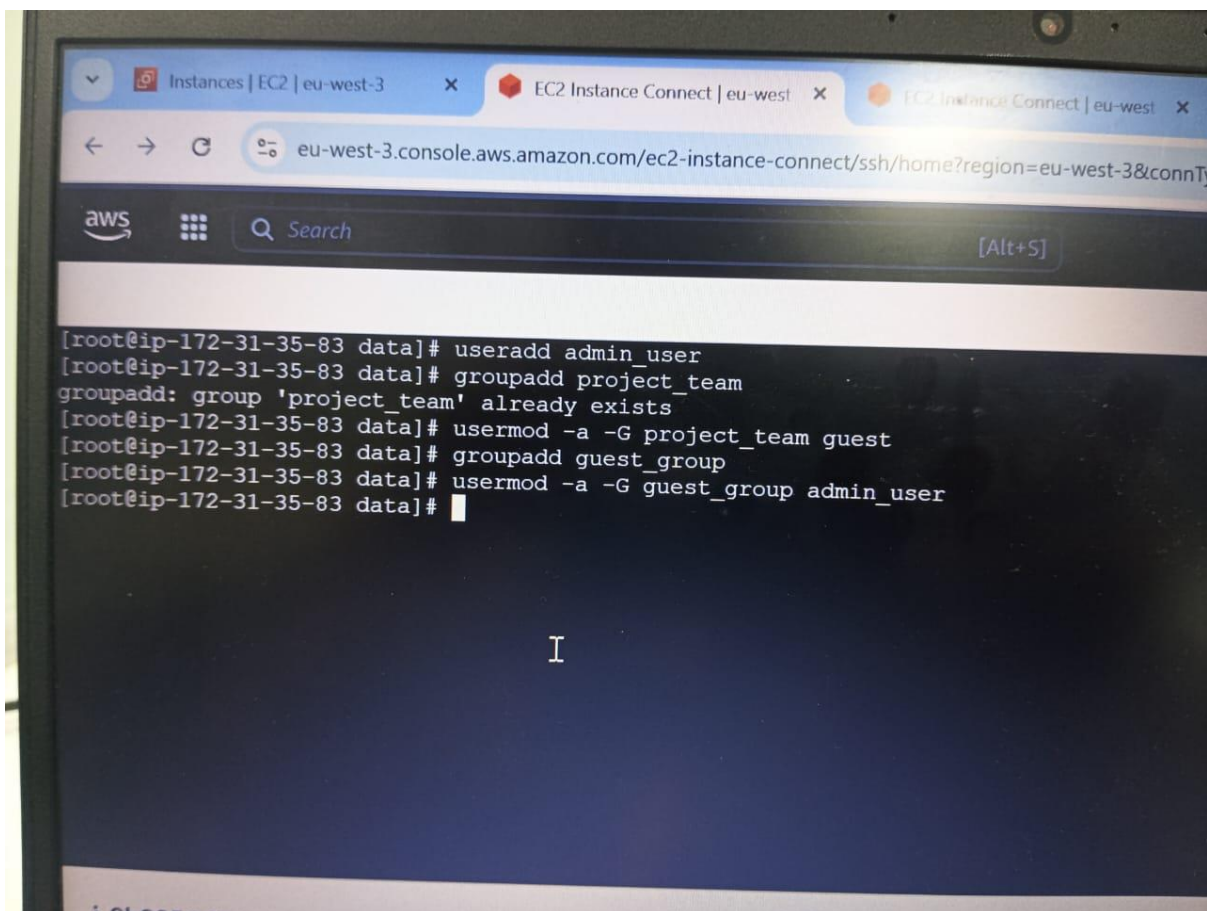
[root@ip-172-31-35-83 data]#
```

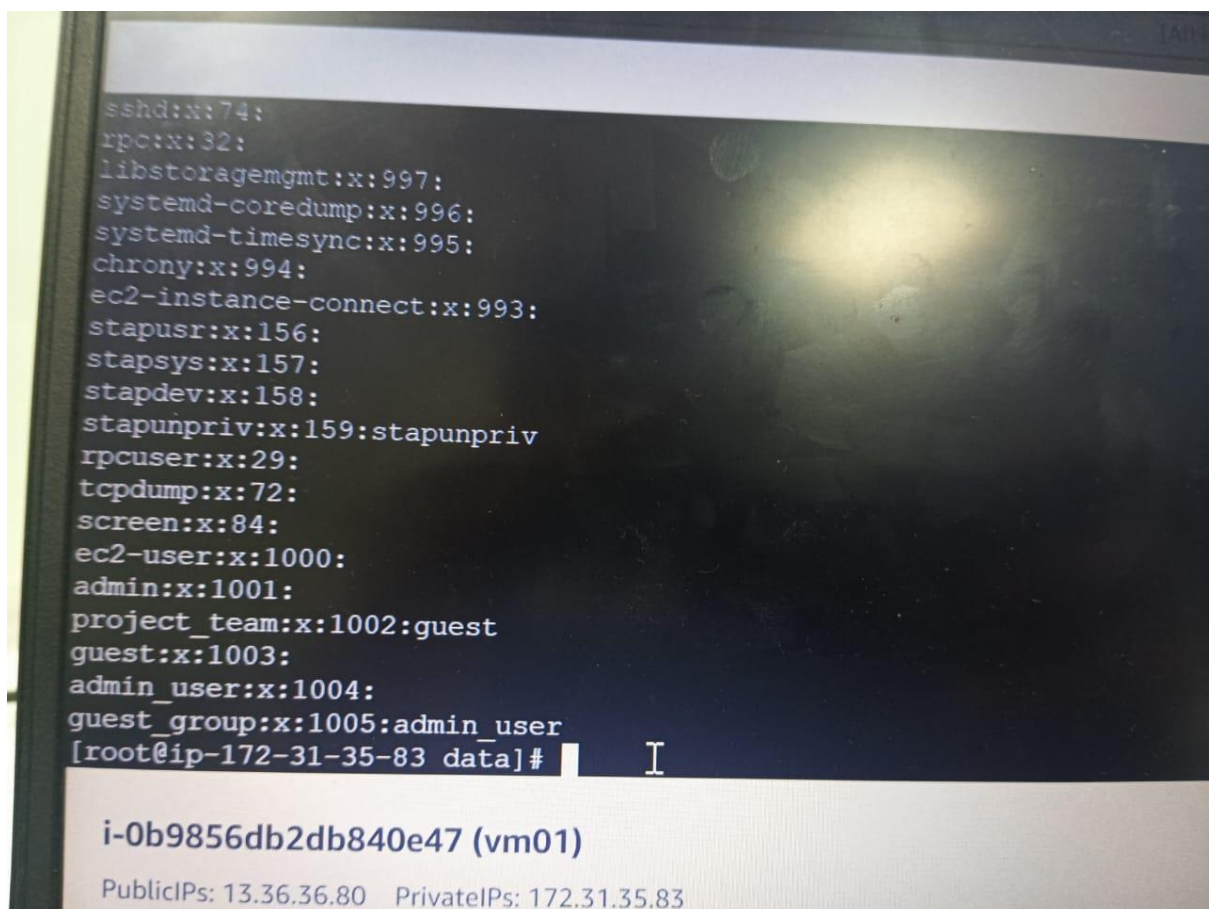
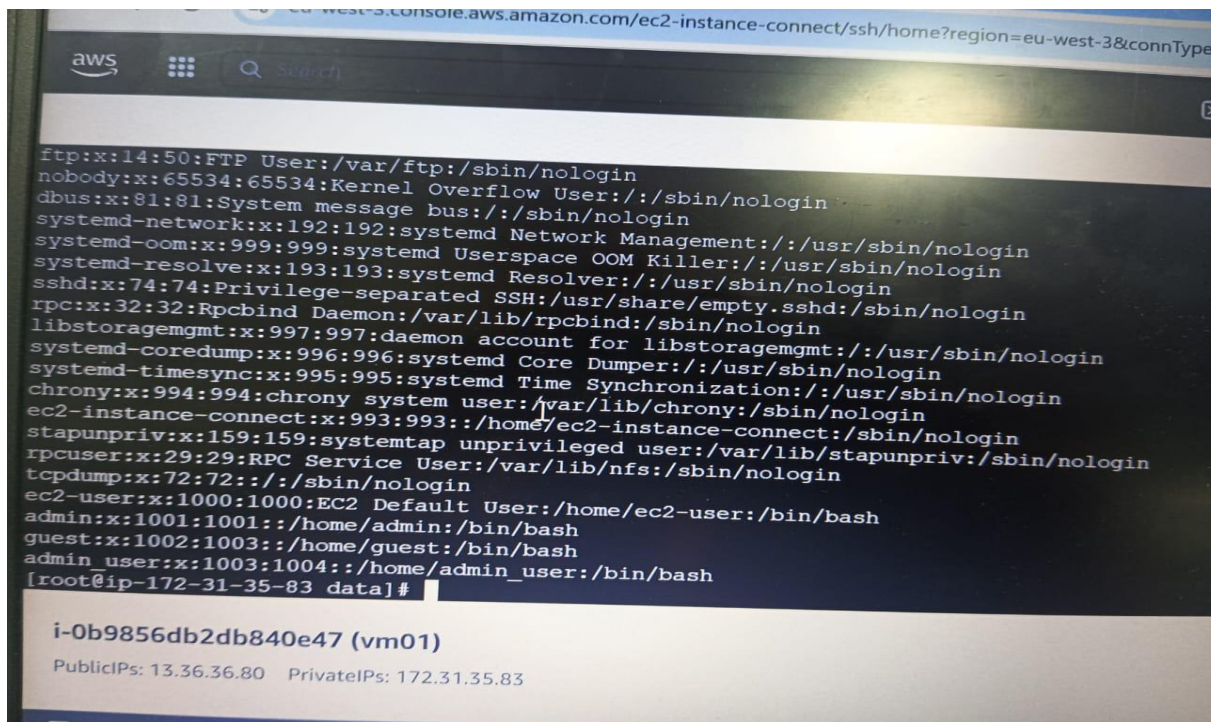
The instance information at the bottom remains the same: **i-0b9856db2db840e47 (vm01)**, PublicIPs: 13.36.36.80, and PrivateIPs: 172.31.35.83. The footer shows the 'CloudShell' label, 'Feedback' link, 'Console Mobile App' button, and a copyright notice '© 2025, Amazon'.



## 5. User and Group Administration (Instance 1)

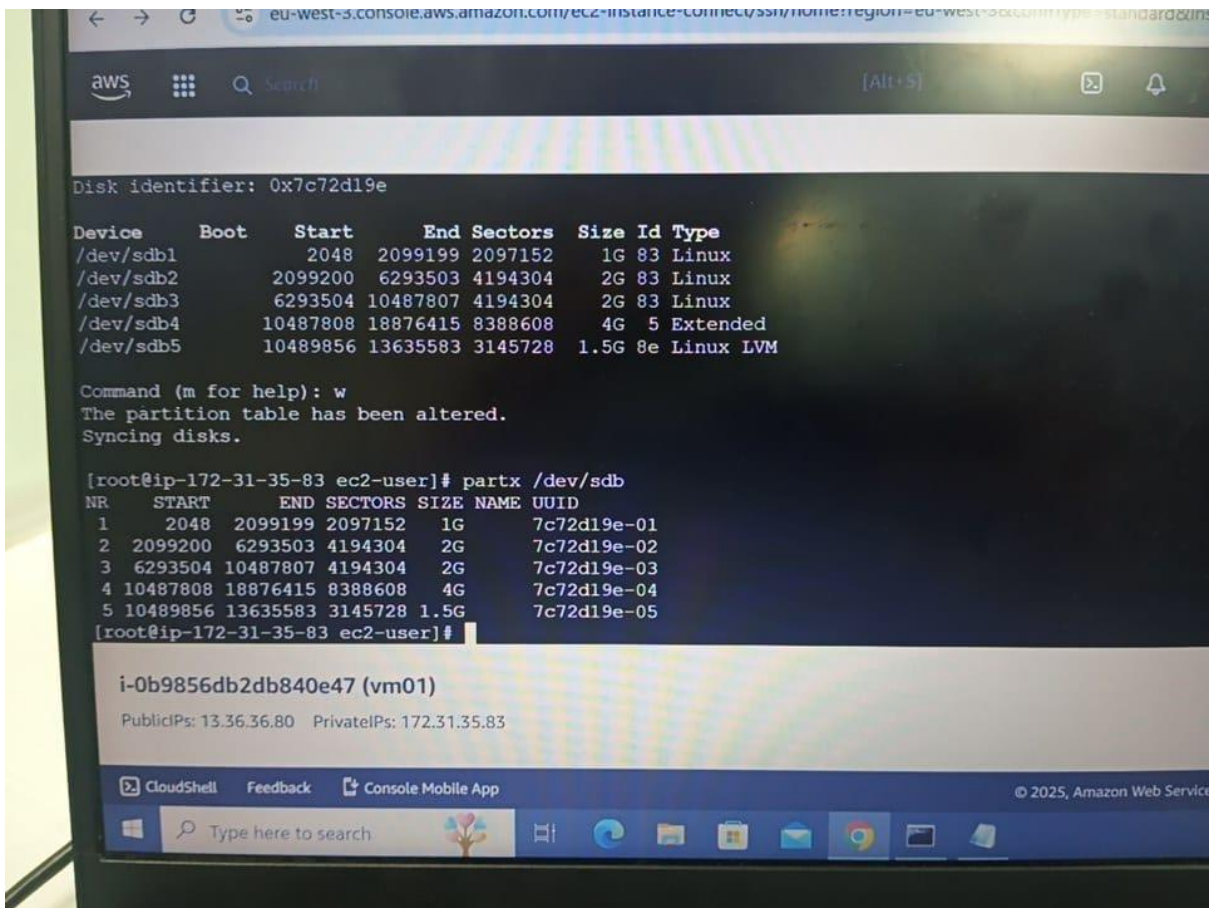
- Create a user named admin\_user.
- Create a group named project\_team.
- Add admin\_user to the project\_team group.
- Create another user named guest\_user and add guest\_user to a new group called guest\_group.
- Verify that the users and groups were created correctly and that the memberships are accurate by using the id and groups commands.





## 6. Partitions and LVM Configuration (Instance 1)

- Check the current partitions and details using lsblk or fdisk -l.
- Create a new partition (if applicable), initialize it as an LVM physical volume, and name the partition /dev/sdb1 (or a similar available device).
- Create a volume group named vg\_data using the new partition.
- Create a logical volume named lv\_data\_backup of size **500MB** within the volume group.
- Format the logical volume with the **xfs** file system and mount it to data\_backup.
- Verify the mounted logical volume using df -h.
- Increase the logical volume size to 1GB.



The screenshot shows an AWS CloudShell terminal window. At the top, the URL bar indicates the console is for an EC2 instance in the eu-west-3 region. The terminal output shows the disk identifier 0x7c72d19e and a table of partitions. Below the table, a message states 'The partition table has been altered. Syncing disks.' followed by the execution of the 'partx /dev/sdb' command. The output of this command shows five partitions with their respective start, end, sectors, size, name, and UUID. At the bottom of the terminal, the instance ID 'i-0b9856db2db840e47 (vm01)' and its IP addresses are displayed. The bottom of the image shows the Windows taskbar with the search bar and several application icons.

```
Disk identifier: 0x7c72d19e

Device Boot Start End Sectors Size Id Type
/dev/sdb1 2048 2099199 2097152 1G 83 Linux
/dev/sdb2 2099200 6293503 4194304 2G 83 Linux
/dev/sdb3 6293504 10487807 4194304 2G 83 Linux
/dev/sdb4 10487808 18876415 8388608 4G 5 Extended
/dev/sdb5 10489856 13635583 3145728 1.5G 8e Linux LVM

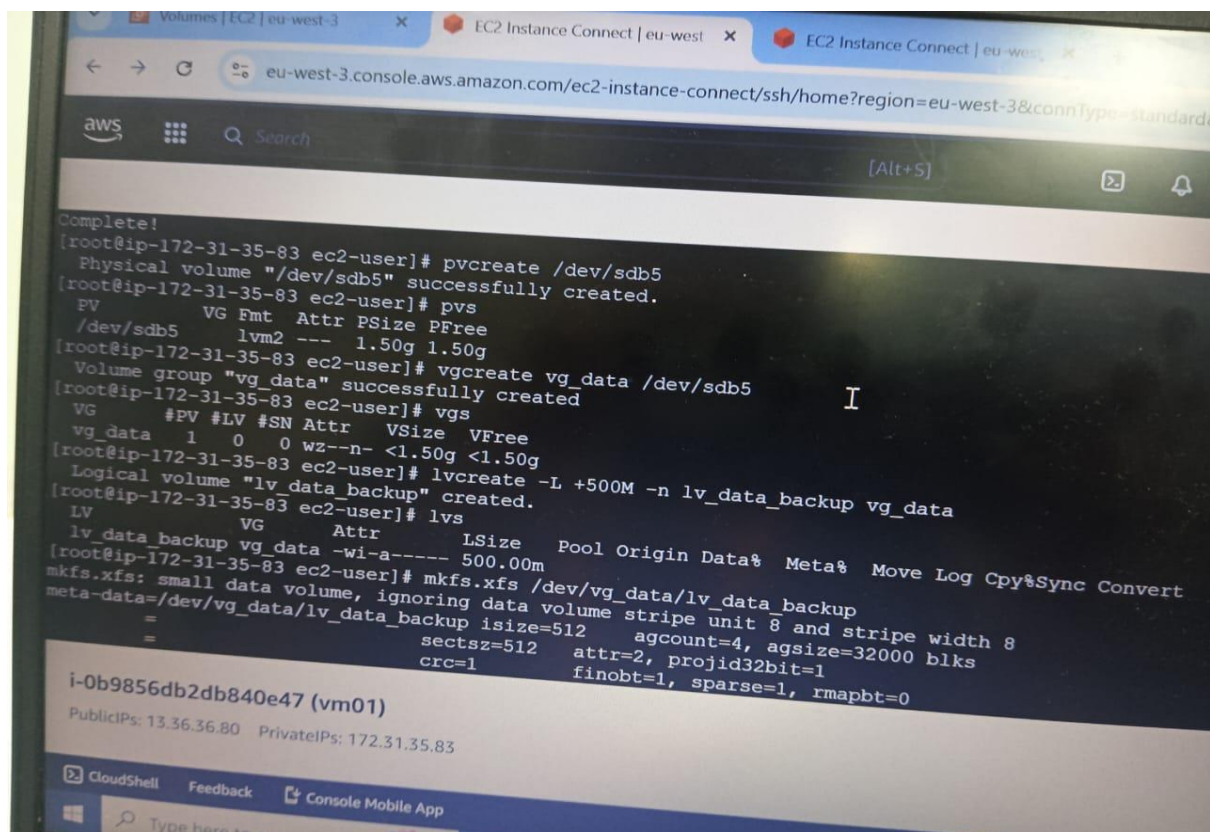
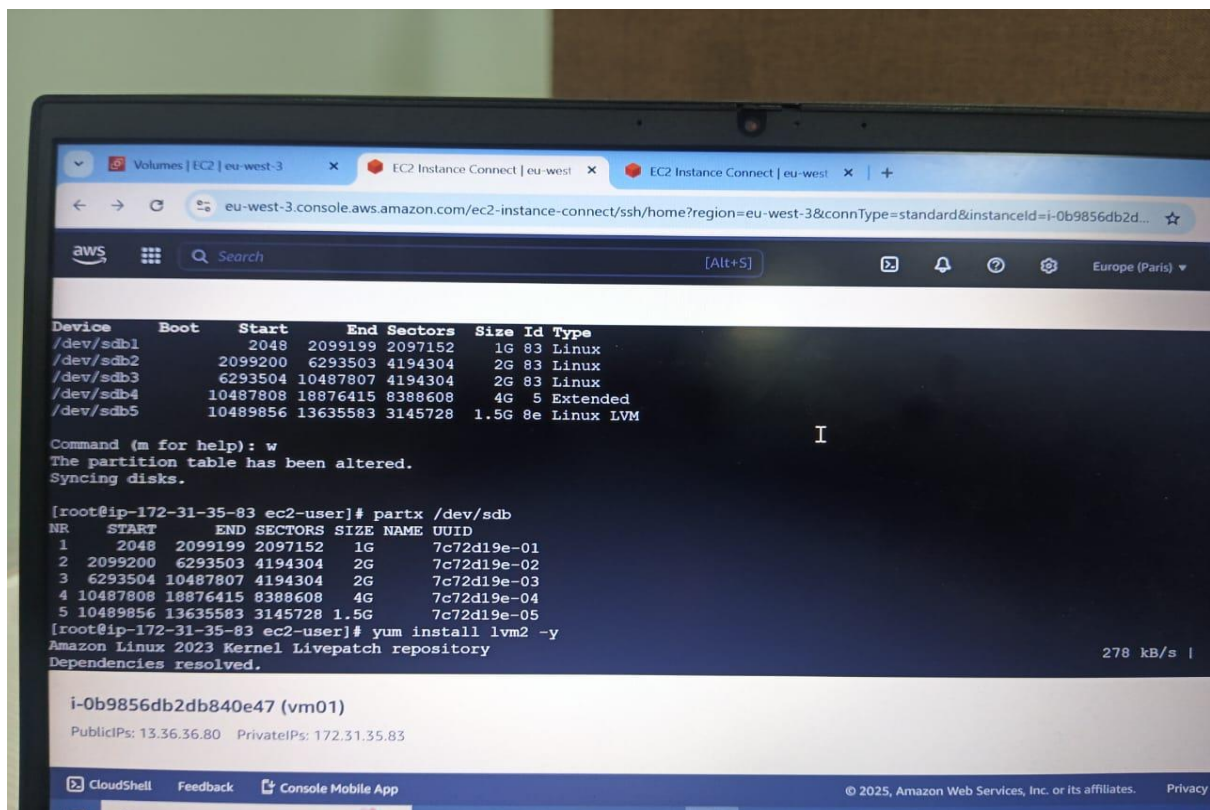
Command (m for help): w
The partition table has been altered.
Syncing disks.

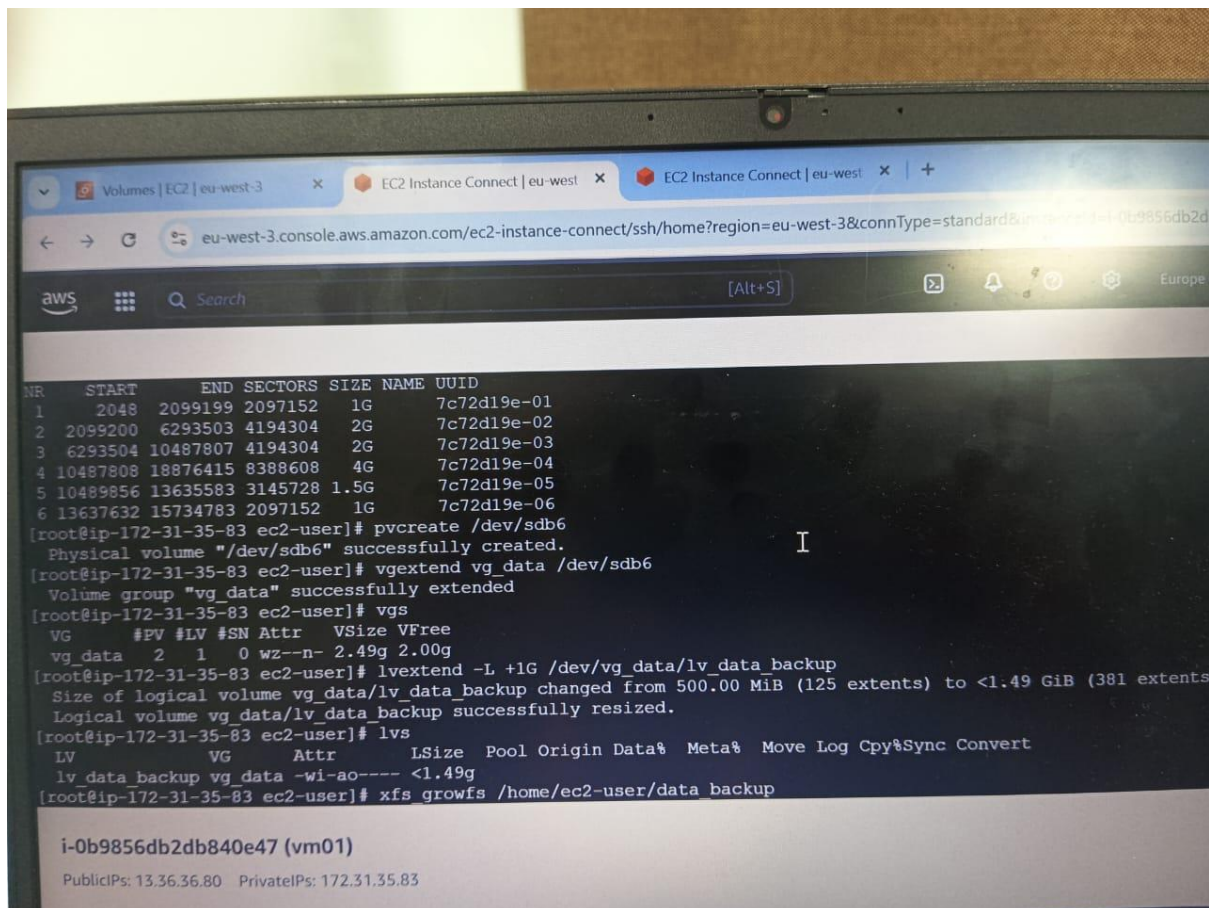
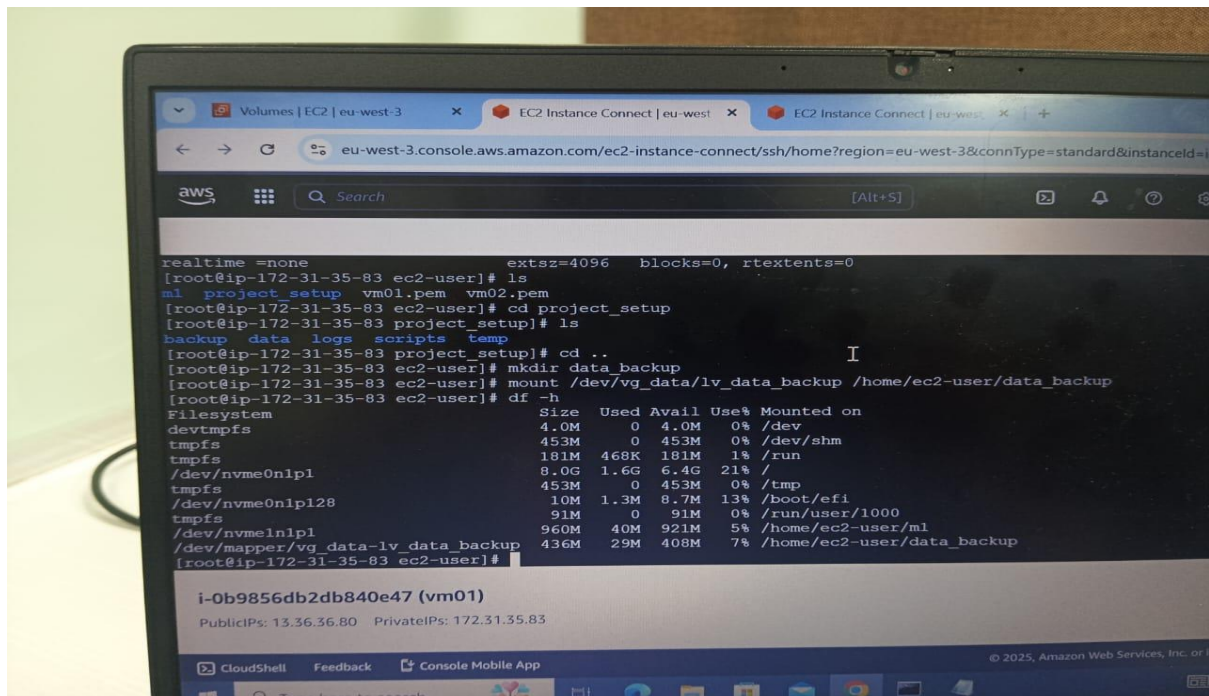
[root@ip-172-31-35-83 ec2-user]# partx /dev/sdb
NR START END SECTORS SIZE NAME UUID
 1 2048 2099199 2097152 1G 7c72d19e-01
 2 2099200 6293503 4194304 2G 7c72d19e-02
 3 6293504 10487807 4194304 2G 7c72d19e-03
 4 10487808 18876415 8388608 4G 7c72d19e-04
 5 10489856 13635583 3145728 1.5G 7c72d19e-05
[root@ip-172-31-35-83 ec2-user]#

i-0b9856db2db840e47 (vm01)
PublicIPs: 13.36.36.80 PrivateIPs: 172.31.35.83

CloudShell Feedback Console Mobile App
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```









The screenshot shows an AWS CloudShell terminal window. The top part of the terminal displays the output of the `df -h` command, showing disk usage for various filesystems. Below this, the output of the `cat /etc/passwd` command is shown, displaying user information for `root` and `ec2-user`. The terminal also shows the output of the `df -h` command again, and the `cat /etc/passwd` command. The bottom part of the terminal shows the output of the `df -h` command, and the `cat /etc/passwd` command. The terminal window has a dark background and a light-colored text.

```
= reflink=1 bigtime=1 inobtcount=1 nnext64=0
= exchange=0
data = bsize=4096 blocks=128000, imaxpct=25
= sunit=0 swidth=0 blks
naming =version 2 bsize=4096 ascii-ci=0, ftype=1, parent=0
log =internal log bsize=4096 blocks=16384, version=2
= sectsz=512 sunit=0 blks, lazy-count=1
realtime =none extsz=4096 blocks=0, rtextents=0
data blocks changed from 128000 to 390144
[root@ip-172-31-35-83 ec2-user]# df -h
Filesystem Size Used Avail Use% Mounted on
devtmpfs 4.0M 0 4.0M 0% /dev
tmpfs 453M 0 453M 0% /dev/shm
tmpfs 181M 472K 181M 1% /run
/dev/nvme0n1p1 8.0G 1.6G 6.4G 21% /
tmpfs 453M 0 453M 0% /tmp
/dev/nvme0n1p128 10M 1.3M 8.7M 13% /boot/efi
tmpfs 91M 0 91M 0% /run/user/1000
/dev/nvme1n1p1 960M 40M 921M 5% /home/ec2-user/m1
/dev/mapper/vg_data-lv_data_backup 1.5G 36M 1.4G 3% /home/ec2-user/data_backup
[root@ip-172-31-35-83 ec2-user]#
```

i-0b9856db2db840e47 (vm01)  
PublicIPs: 13.36.36.80 PrivateIPs: 172.31.35.83

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## 7. Archiving and Compression (Instance 1)

- Create a **tarball** backup of the `project_setup` directory
- Verify the contents of the tarball with `tar -tf project_setup_backup.tar.gz`.
- Extract the tarball to a new directory called `extracted_project_setup`:



```
eu-west-3.console.aws.amazon.com/ec2-instance-connect/ssh/home?region=eu-west-3&cor

aws

[root@ip-172-31-35-83 ec2-user]# ls
data_backup ml project_setup vm01.pem vm02.pem
[root@ip-172-31-35-83 ec2-user]# tar -cvf project_setup.tar project_setup
project_setup/
project_setup/data/
project_setup/data/overview.txt
project_setup/data/configuration.txt
project_setup/data/summary.txt
project_setup/data/analysis_backup.txt
project_setup/data/archive.txt
project_setup/backup/
project_setup/backup/overview.txt
project_setup/logs/
project_setup/logs/report.txt
project_setup/temp/
project_setup/scripts/
[root@ip-172-31-35-83 ec2-user]# ls
data_backup ml project_setup project_setup.tar vm01.pem vm02.pem
[root@ip-172-31-35-83 ec2-user]#
```

i-0b9856db2db840e47 (vm01)  
PublicIPs: 13.36.36.80 PrivateIPs: 172.31.35.83

```
Instances | EC2 | eu-west-3
EC2 Instance Connect | eu-west-3
eu-west-3.console.aws.amazon.com/ec2-instance-connect/ssh/home?region=eu-west-3&connType=standard&instanceId=i-0b9856db2db840e47

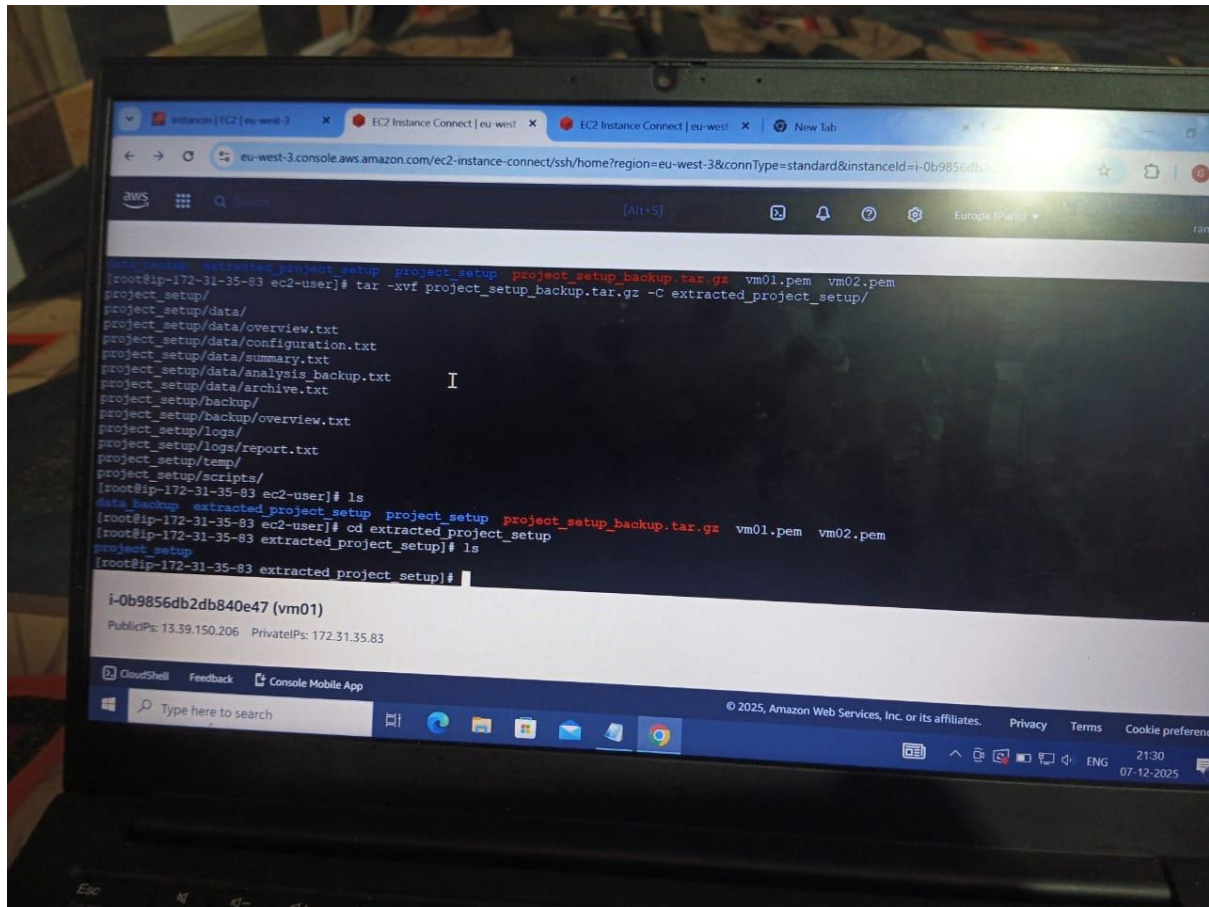
aws

tar: Exiting with failure status due to previous errors
[root@ip-172-31-35-83 ec2-user]# ls
data_backup project_setup project_setup_backup.tar.gz vm01.pem vm02.pem
[root@ip-172-31-35-83 ec2-user]# cd data_backup
[root@ip-172-31-35-83 data_backup]# ls
[root@ip-172-31-35-83 data_backup]# cd ..
[root@ip-172-31-35-83 ec2-user]# tar -tf project_setup_backup.tar.gz
project_setup/
project_setup/data/
project_setup/data/overview.txt
project_setup/data/configuration.txt
project_setup/data/summary.txt
project_setup/data/analysis_backup.txt
project_setup/data/archive.txt
project_setup/backup/
project_setup/backup/overview.txt
project_setup/logs/
project_setup/logs/report.txt
project_setup/temp/
project_setup/scripts/
[root@ip-172-31-35-83 ec2-user]#
```

i-0b9856db2db840e47 (vm01)  
PublicIPs: 13.39.150.206 PrivateIPs: 172.31.35.83

CloudShell Feedback Console Mobile App

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## 9. File Transfer Between Instances Using SCP

- From Instance 1, use **SCP** to copy the project\_setup\_backup.tar.gz file to Instance 2:
- SSH into Instance 2 and verify the transfer by listing the file:

